ZAMBIA UNIVERSITY COLLEGE OF TECHNOLOGY

ICT 2201 Database Systems

Final examination - Academic year 2023, semester 2

Time allowed: 3 hours Total marks: 100 Lecturer: Dr M. Sinkala

Instructions: answer ALL questions. Write clearly. Show all working.

Question 1: ER modelling

(a) In an ER diagram, a weak entity is one that:

(b) A many-to-many relationship is implemented in a relational schema by:

(c) An attribute that can be calculated from other attributes is:

(d) Cardinality in an ER relationship describes:

Question 2: Relational model

(a) In the relational model, a row of a table is formally called a:

(b) Which key uniquely identifies each row and cannot be NULL?

(c) A foreign key enforces which kind of integrity?

(d) The set of allowed values for an attribute is its:

Question 3: SQL

(a) Which SQL clause filters groups after aggregation?

(b) Which statement removes all rows but keeps the table structure, and cannot be rolled back i
n most systems?

(c) A LEFT JOIN returns:

(d) Which aggregate ignores NULL values?

Question 4: Transactions

(a) The A in ACID stands for:

(b) Which property ensures committed data survives a crash?

(c) A dirty read happens when a transaction reads data that is:

(d) Two transactions each waiting for a lock the other holds is called: